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Modeling multi-target defender-attacker games with quantal response attack strategies

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ABSTRACT

Due to the growing interest in incorporating behavioral models within the defender-attacker game literature, the perfect rationality assumption of adversaries has been relaxed in many recent works. This paper studies four multi-target defensive resource allocation in defender-attacker games by using a quantal response model, in which the attacker might be bounded rational. We compare the results from the proposed models with the traditional game theoretic model where the attacker behaves rationally. Moreover, we studied the impacts of the attacker's rationality level on the defender's optimal defense strategies. Real data from the Urban Areas Security Initiative (UASI) and the Global Terrorism Database (GTD) are employed to estimate the rationality parameter as well as numerically illustrate the model. The results show that: (a) in the quantal response model, the defender's expected loss at equilibrium increases with the attacker's rationality parameter and the expected loss approach the results from the game theoretic model; (b) when the attacker's rationality parameter in the quantal response model approaches infinity, both the equilibrium defensive allocation and expected loss approach to the results from the game theoretic model; (c) when the attacker uses quantal response while the defender falsely believes that the attacker uses game theoretic response, the expected loss will be similar to the model where both players utilize game theoretic response, but higher than the expected loss from the model where both players use quantal response; (d) when the attacker uses game theoretic response while the defender falsely believes that the attacker uses quantal response, the expected equilibrium loss will be higher than the ones from the other three scenarios; and (e) the quantal response model gives a closer resource allocation to the real allocation than the game theoretic model does. This paper provides some novel insights to defensive resource allocation when considering that the attacker is bounded rationality.

1. Introduction

Terrorism is one of the most serious concerns to homeland security. Since the terrorism attacks on September 11th, 2001, hundreds of billions of dollars have been spent on homeland security as shown in Fig. 1. However, the risk-related measurements of resource allocation are still limited [1].

Game theory has been widely applied to study homeland security issues, in which results have shown that defender-attacker games are effective in studying the counter-terrorism defensive resource allocation problems [3–6]. Levitin and Hausken [7] consider the scenario of deploying false targets to protect a single object as part of a defense strategy. Golalikhani and Zhuang [8] propose a game theoretic model to assign the defense resource to any subset targets with the functional and geographical similarities. Levitin and Hausken [9] study the influence of the probability that an attacker can identify all elements in a system using a non-cooperative two-period minmax game between the defender and the attacker. Similarly, a system consisting of identical elements, and each element can be protected individually is studied in Levitin et al. [10]. A two-period game between a government and a terrorist is studied in Hausken and Zhuang [11], in which the terrorist could stockpile attack resources from the first to the second period. Hausken and Zhuang [12] study the timing and deterrence of terrorist attacks in a T-period game where one attack happens in each period. Hausken and He [13] develop a game-theoretic model for protecting an infrastructure of N targets to determine optimal resource investments across vulnerable targets against terrorism threats. Multiple-period, multi-target attacker-defender game has been studied in Shan and Zhuang [14]. Multiple attacking options of the attacker in multi-target attacker-defender games are studied in Zhang and Zhuang [15]. Ben Yaghlane et al. [16] define the concept of network survivability in the context of defense/attack strategies, and investigate the Nash

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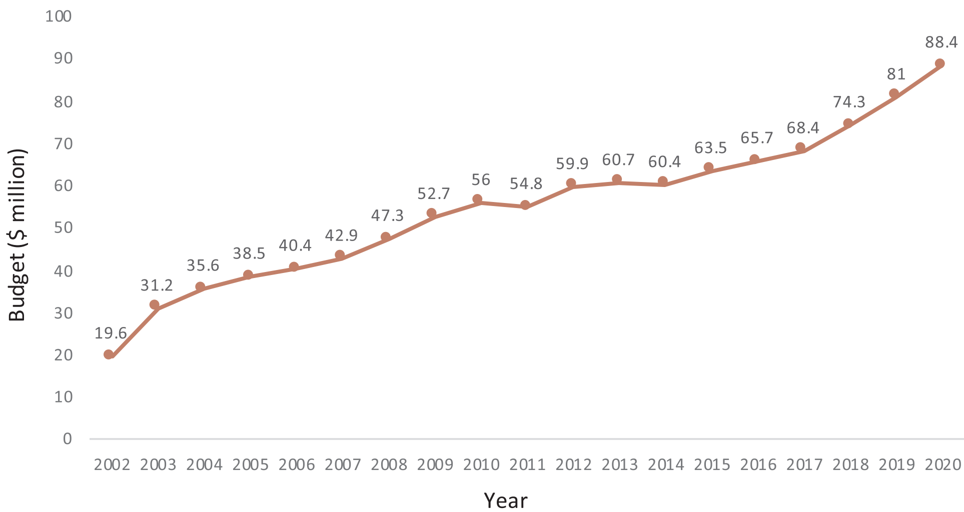

Fig. 1. Annual budget requests for the U.S. Department of Homeland Security. Source: [2].

equilibrium. Shan and Zhuang [17] studied attacker-defender games in the context of smart grids. Peng et al. [18] considers the optimal allocation of defense resources to defend a parallel system, and disinformation actions are studied for both the defender and the attacker. Zhai et al. [19] studied asymmetric player utilities in attacker-defender games. Aziz et al. [20] studied an attacker-defender game considering substitution and complementary effects of defense investments. Xu and Zhuang [21] studied a sequential one-defender-N-attacker game. As discussed above, various scenarios have been studied and applied to different systems. However, these research study the defensive recourse allocation problem by finding the Nash equilibrium of the model. And none of them consider the cognitive limitations of human mind when making decisions, leaving the bunded rationality of the players unstudied. In this paper, we model the homeland security resource allocation problem considering multiple targets and study how the bunded rationality of the players would impact the results.

Risk preferences have been studied in risk modeling [22–24]. In defensive game-theoretic models, Zhang et al. [25] model and analyze equilibrium results for a sequential defender-attacker game, where the defender defends multiple potential targets while considering the risk preferences of the attacker and defender, and study how risk preferences would affect a player's behavior in equilibrium. Wu et al. [26] study the optimal defense policy for an object with defense measures, where the player's risk preferences are introduced through cumulative prospect theory. The solution to those types of models is still a Nash equilibrium for simultaneous-move games and a subgame-perfect Nash equilibrium for sequential games. However, players in practice have noted that they do not behave as the Nash equilibrium predicts [23]. Game theory has been criticized to attributing excessive levels of rationality, knowledge, and computational ability to potential terrorists. The assumption of the traditional game theoretic model, that all players are rational, makes the models sometimes inefficient for application in practice. The attacker may be bounded rational but choose non-optimal targets in a game theoretic sense, which may invalidate many game theoretic models developed in the recent decade [27]. In this paper, quantal response that considers the non-optimal selections is studied to address this problem.

Due to the bounded rationality of human behaviors, the assumption that the attacker is perfectly rational which has been adopted in most researches in literature should be relaxed. Researchers have made significant effort to construct utility functions that could model human's decision making behavior. Alternative theories of decision making have been constructed in an effort to improve more realistically model decision making. These types of theories include regret theory [28], security-protection/aspiration theory [29,30], and cumulative prospect theory [31,32]. Quantal response equilibrium (QRE) is a prominent approach for studying bounded rationality [33,34]. QRE is an approach which assumes that the players observe the reward for any particular action with some error, and that there exist errors in human decision making. Instead of strictly maximizing utility, individuals respond probabilistically (instead of deterministically) in games [33]. Quantal response has been employed in many fields, such as biology and medical science [35], private-value auction [36], bioassay [37], and animal carcinogenicity experiments [38]. Additionally, quantal response has not been extensively used in defender-attacker games to study the defensive resource allocation.

Due to the nature of human behavior, it is critical that adversaries' bounded rationality is addressed when computing the optimal equilibrium in the defender-attacker games. Pita et al. [39] investigate how to build algorithms for agent interactions with human adversaries by addressing boundedly rational opponents and anchoring biases those opponents may have. Yang et al. [40] incorporate rank-related expected utility into quantal response by considering the fact that individuals are attracted to extreme events. This paper employs quantal response to model attacker bounded rationality in game theoretic settings. We propose a new multi-target resource allocation quantal response model and compare that with the traditional game theoretic models, utilizing real-world data illustration.

The remainder of this paper is structured as follows. Section 2 introduces the notation and four models; Section 3 discusses the analytical results and gives propositions; Section 4 introduces the data that is used to estimate the parameters and perform numerical illustration; Section 5 provides numerical illustrations; Section 6 concludes the paper. Appendices provide proofs for the propositions.

2. Model formulation

2.1. Notation

Table 1 presents all notations utilized in this paper. Among them, n is the number of targets, and i is the index of the target. The total defense budget is C, and v_i represents the value of target i, for $i = 1, 2, \ldots, n$. The notation c_i refers to the budget allocation for the target i, and $L(\mathbf{c}, \mathbf{h})$ refers to total expected loss. The notation $h_i(\mathbf{c})$ represents the attack probability for the attacker, while $p_i(c_i)$ represents the success probability of an attack. In this paper, we let $p_i = \frac{a}{a+c_i}$, where a is a constant, and $a > 0$. The value of a affects the effectiveness of the budget allocation. If $a \to \infty$, then the success probability is 1, and the effectiveness of the budget allocation is low; if $a \to 0$, the effectiveness of the budget allocation is high. The notation λ is the quantal

Table 1
Notation used in the paper.

Parameters	
$i = 1,...,n$	Target index, n is number of targets
N	Observed number of attacks historically
C	Total defender's budget
$p_i(c_i)$	Success probability of an attack on target i
v_i	Valuation of target i
J	The set of targets with positive attack probabilities. That is $J \equiv \{i: h_i(\mathbf{c}) > 0;\ i = 1, 2, ..., n\}$
λ	The level of rationality in the attacker's response
Functions	
$L(c, h)$	Total expected loss
Decision Variables	
c_i	Budget allocated to target i, $\sum_{i=1}^{n} c_i \le C$
h_i	Probability of a terrorist attack on target i for quantal response model
$\mathbf{c} = [c_1, ..., c_n]$	Vector of the defender's resource allocation
$\mathbf{h} = [h_1, ..., h_n]$	Vector of the attacker's attack probability
Four Possible Losses	
L_{GG}	Expected loss when both defender and attacker consider game theoretic response
L_{GQ}	Expected loss when defender considers game game theoretic response, but attacker uses quantal response model
L_{QG}	Expected loss when defender considers quantal response model, but attacker uses game game theoretic response
L_{QQ}	Expected loss when both defender and attacker consider quantal response model

response parameter representing the attacker's rationality level. As $\lambda \to 0$, the attacker is completely irrational and selects the target uniformly. As $\lambda \to \infty$, the attacker is perfectly rational and selects the target with the highest expected loss. The last four notations (L_{GG}, L_{GQ}, L_{QG} and L_{QQ}) are expected losses from four different models when considering two possible (game theoretic and quantal) responses, where the defender may not know which response the attacker will adopt. Thus, they may have false beliefs for the attacker. This will be discussed in Section 2.2.

2.2. The models

To identify the optimal defensive budget allocation, a sequential game similar to the one from [41] is utilized. We acknowledge that there may be multiple rounds in the resource allocation in practice, in this paper we only consider one period game. Related multiple-period attacker-defender game are studied in the literature, such as Zhang and Zhuang [15], Jose and Zhuang [42], Shan and Zhuang [43]. In this allocation, the defender moves first and minimizes the total expected loss subject to budget constraints:

$$\begin{aligned} \min L(\mathbf{c}, \mathbf{h}) \ &= \textstyle\sum_{i=1}^{n} h_i(\mathbf{c}) p_i(c_i) v_i \\ \text{subject to} \ &: c_1 + c_2 + \ldots + c_n \le C. \end{aligned} \tag{1}$$

where $h_i(\mathbf{c})$ is the attack probability to target i given the allocation c; the success probability of the attack to target i is represented as $p_i(c_i)$, and v_i is the valuation of target i. We can consider $h_i(c)$, $p_i(c_i)$ and v_i as threat, vulnerability and consequence, respectively.

There are two possible responses for the attacker, quantal response and game theoretic response. The defender may correctly or falsely believe that the attacker's response functions. Therefore, there may exist four different situations as introduced in the following four subsections.

2.2.1. Pure quantal response

In this model, the defender correctly believes that the attacker uses quantal response. Quantal response equilibrium, first introduced by Richard McKelvey and Thomas Palfrey in McKelvey and Palfrey [33], is a concept from behavioral game theory. In quantal response equilibrium (QRE), the players do not always choose the best response as in Nash Equilibrium, instead the higher expected payoffs are chosen with higher probability. QRE provides an equilibrium notion with bunded rationality. The players are modeled as æbetter respondersg rather than best responders [44]. Although this case analogizes to the QRE, the non-rational noise is only added to the response function of the attacker. The defender will select the strategy that minimizes the total expected loss.

In practice, the logistic quantal response function is the most commonly adopted in QRE [33]. The probability $h_i(\mathbf{c})$ of a terrorist attack on target i, for $i = 1, ..., n$, is denoted as

$$h_i(\mathbf{c}) = \frac{e^{\lambda p_i(c_i) v_i}}{\sum_{i=1}^{n} e^{\lambda p_i(c_i) v_i}} \tag{2}$$

where λ is the attacker's rationality parameter, and we take the λ in quantal response Equation (2) as the level of rationality in the attacker's response [33]. The extreme cases are: (1) when $\lambda = 0$, the attacker is fully random, and chooses to attack each target with equal probability $\frac{1}{n}$; (2) when $\lambda \to \infty$, the attacker is fully strategic, and would choose the target with the largest expect loss to attack.

Inserting the attack probability function (2) to the defender objective function (1), the equilibrium allocation and attacking probability $(\mathbf{c}^*, \mathbf{h}^*)$ could be calculated, where

$$\mathbf{c}^* = \arg\min_{\mathbf{c}} L(\mathbf{c}, h(\mathbf{c})) \tag{3}$$

and

$$\mathbf{h}^* = h(\mathbf{c}^*) \tag{4}$$

The defender's corresponding expected loss is denoted as L_{QQ}, which is the expected loss when both defender and attacker considers quantal response model:

$$L_{QQ} = L(\mathbf{c}^*, \mathbf{h}^*) = L(\mathbf{c}^*, \mathbf{h}(\mathbf{c}^*)) \tag{5}$$

2.2.2. Pure game theoretic response

In this model, the attacker uses game theoretic response, and the defender correctly believes so. This case is the traditional defender-attacker game, where a Subgame Perfect Nash Equilibrium (SPNE) is utilized. The attacker's objective is to maximize the total expected loss. Therefore, a best response for attacking is as follows:

$$\hat{h}_i(\mathbf{c}) = \begin{cases} \frac{1}{|J|}, & \text{if } i \in J \\ 0, & \text{if } i \notin J \end{cases} \tag{6}$$

where $J \equiv \{j: p_j(c_j)v_j = \max_i p_i(c_i)v_i\}$ refers to the set of targets with the highest expected damage. The attacker adapts his attack strategy based on the resource allocation of the defender. Equation (6) shows the best response of the attacker after observing the defenders allocation. The attacker chooses to attack the targets in J with the same probability [25,45]. For the defender, the objective is to minimize the total expected loss by considering the best response for the attacker $\hat{\mathbf{h}}(\mathbf{c})$. Therefore, according to Equation (6), we have

$$\begin{aligned} \min_{\mathbf{c}} L(\mathbf{c}, \hat{\mathbf{h}}(\mathbf{c})) \ &= \textstyle\sum_{i=1}^{n} \hat{h}_i(\mathbf{c}) p_i(c_i) v_i \\ \text{subject to} \ &: c_1 + c_2 + \ldots + c_n \le C \end{aligned} \tag{7}$$

In this model, we call a pair of strategies, $(\mathbf{c}^{**}, \hat{\mathbf{h}}^*)$, a Subgame Perfect Nash Equilibrium (SPNE) for the sequential defender-attacker game, where

$$\mathbf{c}^{**} = \arg\min_{\mathbf{c}} L(\mathbf{c}, \hat{h}(\mathbf{c})) \tag{8}$$

$$\hat{\mathbf{h}}^* = \hat{\mathbf{h}}(\mathbf{c}^{**}) \tag{9}$$

In this case, the defender's total expected loss will be

$$L_{GG} = L(\mathbf{c}^{**}, \hat{\mathbf{h}}^{*}) = L(\mathbf{c}^{**}, \hat{\mathbf{h}}(\mathbf{c}^{**})) \tag{10}$$

2.2.3. Falsely believing game theoretic response

In this situation, the defender allocates resource by falsely considering that the attacker is using game theoretic response, while the attacker uses quantal response. Therefore, we have

$$L_{GQ} = L(\mathbf{c}^{**}, \mathbf{h}^{*}) \tag{11}$$

where $\mathbf{c}^{**}$ is defined in Eq. (8), and $\mathbf{h}^{*}$ is defined in Eq. (4).

2.2.4. Falsely believing quantal response

In this scenario, the defender falsely believes that the attacker is using quantal response, when the attacker is actually using the game theoretic model. The attacker's attacking probability is based on the best response of the game theoretic model $\hat{\mathbf{h}}(\mathbf{c})$ as specified in Equation (6). However, the optimal allocation from defenders is changed to that of quantal response model $\mathbf{c}^{*}$, as defined in Eq. (3). The defenders' expected loss in this case becomes

$$L_{QG} = L(\mathbf{c}^{*}, \hat{\mathbf{h}}(\mathbf{c}^{*})) \tag{12}$$

2.3. Summary of the four models

In summary, there exist four scenarios as shown in Table 2.

3. Analytical results

PROPOSITION 1. Quantal response equilibrium $(\mathbf{c}^{*}, \mathbf{h}(\mathbf{c}^{*}))$ exists in the model defined by Eqs. (1) and (2). When the rationality parameter λ approaches to infinity, the attack probability will converge to the one from the game theoretic model i.e.,;

$$\lim_{\lambda\to\infty}\mathbf{h}(\mathbf{c}^{*}) = \hat{\mathbf{h}}(\mathbf{c}^{**}) \tag{13}$$

Remark: When λ increases, the attacker becomes more rational, which gives him a higher probability to attack target i with the highest value $p_i v_i$. Thus, when λ approaches to infinity, the attacker becomes fully rational and will act like that of a game theoretic model. In this quantal response case, the defender should gather information to best estimate the rationality of the attacker in order to make the effective resource allocation.

PROPOSITION 2. When considering game theoretic response, if $c_i > 0$, the equilibrium defensive allocation c satisfies the condition $c_i = \frac{v_i(na + C)}{\sum_{i=1}^{n} v_i} - a$ where $p_1v_1 = p_2v_2 = ... = p_nv_n$. No matter what response the attacker uses, the total expected loss always equals to $p_i v_i$.

Remark: When $c_i > 0$, the budget C is sufficiently large to make all the targets receive some allocation. Since the targets have the same expected loss, the attacker is indifferent on which to specifically target. The attacker will randomly choose one to attack. Hence in this game theoretic response case, the defender should strategically allocate resource to the more attractive targets to make the targets have the same expected loss.

Table 2
Four scenarios based on the response of the attacker and the beliefs of the defender.

Defender's belief\Attacker's actual response	Game theoretic response	Quantal response
Game theoretic response	L_{GG}=$L(c^{**}, \hat{h}(c^{**}))$	$L_{GQ} = L(c^{**}, h(c^{*}))$
Quantal response	$L_{QG} = L(c^{*}, \hat{h}(c^{**}))$	$L_{QQ} = L(c^{*}, h(c^{*}))$

PROPOSITION 3. When the total budget C increases, the expected losses of the four models decrease.

Remark: As C increases, the allocation to each target i will increase. The increasing of C will lead to a lower attack probability $h_i(c)$ and success probability $p_i(c_i)$, which will reduce the expected loss shown in Eq. (1). The defender should try to invest more. No matter what the best response is from the attacker, a large amount of resource would lead to a smaller damage.

4. Data source and parameter estimation

In this paper, we consider the data for the ten urban areas: New York City, Chicago, San Francisco, Washington, D.C., Los Angeles, Philadelphia, Boston, Houston, Newark, and Seattle-Bellevue. These are the top 10 areas that receive allocations from the UASI in 2004 [46]. Section 4.1 gives the expected property loss for each of the areas; Section 4.2 lists the annual allocation to the ten areas in FY 2004–2014; Section 4.3 discusses the terrorism attacks to the ten areas in FY 2004–2014. These data will be used to estimate the parameter λ in Section 4.4, and the numerical illustration in Section 5.

4.1. Target expected losses

Adopted from Willis et al. [47], we take property loss ($million) as the target valuation in this paper. The target valuations for the ten urban areas are: New York City, $v_1 = 413$; Chicago, $v_2 = 115$; San Francisco, $v_3 = 57$; Washington, D.C., $v_4 = 36$; Los Angeles, $v_5 = 34$; Philadelphia, $v_6 = 21$; Boston, $v_7 = 18$; Houston, $v_8 = 11$; Newark, $v_9 = 7.3$; and Seattle-Bellevue, $v_{10} = 6.7$.

4.2. Resource allocation

"The UASI program funds addressed the unique risk driven and capabilities-based planning, organization, equipment, training, and exercise needs of high-threat, high-density Urban Areas... and assists them in building an enhanced and sustainable capacity to prevent, protect against, mitigate, respond to, and recover from acts of terrorism" [48]. Table 3 shows the budget allocation for the ten urban areas in FY 2004–2014. The resource allocation data will be employed as the defender's defense strategy for parameter estimation in Section 4.4.

4.3. Attacks

We collected the attack incidents data from the Global Terrorism Database (GTD) [49]. GTD is maintained by the National Consortium for the Study of Terrorism and Responses to Terrorism (START). The attack frequencies for the ten urban areas from 2004 to 2014 are shown in Table 4. This is a total of 31 attacks, which are numbered from 1 to 31. The frequency data for terrorism attacks according to the GTD will be taken as the attacker's choice for the parameter estimation in Section 4.4.

4.4. Parameter estimation

We estimate the level of the attacker's rationality λ in this section using the data discussed in Sections 4.1, 4.2, and 4.3. We adopt Maximum Likelihood Estimation to estimate λ in Equation (2). We let $a = 1$ in the success probability function. Given the defender's resource allocation c provided in Table 3 in Section 4.2 and the attacks of the attacker provided in Table 4 in Section 4.3, the likelihood function is

$$L(\lambda) = \prod_{k=1}^{N} h_k(\mathbf{c}; \lambda) \tag{14}$$

where N is the total number of attacks in Table 4. We define $l(\lambda)$ as the

Table 3
UASI allocation for the ten urban areas FY 2004–2014 ($ million).

Urban Area	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
New York City	47	207.6	124.5	134.1	144.2	145.6	151.6	151.6	151.6	174.3	178.9
Chicago	34.1	45	52.3	47.3	45.9	52.3	54.7	54.7	47.7	67.7	69.5
San Francisco	26.5	20.4	28.3	34.1	37.2	40.6	42.8	42.8	26.4	27.3	27.4
Washington, D.C.	29.3	77.5	46.5	61.7	59.8	58	59.4	59.4	51.8	51.8	53
Los Angeles	40.4	69.2	80.6	72.6	70.4	68.3	70	70	61	65.9	67.5
Philadelphia	23.1	22.8	19.5	18.7	18.1	18	23.3	23.3	14.3	17.6	18.5
Boston	19.1	26	18.2	14.2	13.8	14.6	18.9	18.9	10.9	17.6	18
Houston	20	18.6	16.7	25	37.5	39.6	41.5	41.5	23.9	23.9	24
Newark	15.1	12.4	0	0	0	0	0	0	0	0	0
Seattle-Bellevue	16.5	11.8	9.2	10.7	10.3	11.3	11.1	8	4.4	5.5	5.5
Total	271.1	511.3	395.8	418.4	437.2	448.3	473.3	470.2	391.93	451.6	462.3

Source: DHS [48].

log-likelihood function. Therefore the log of the likelihood function of (14) is

$$l(\lambda) = \log[L(\lambda)] = \sum_{k=1}^{N} \ln(h_k(\mathbf{c};\lambda)) \tag{15}$$

We then write the solution formally as the argument λ that maximizes the log-likelihood

$$\hat{\lambda} = \underset{\lambda}{\operatorname{argmax}}\, l(\lambda) \tag{16}$$

According to Table 4, there is a total of 31 attacks in year 2004–2014 for the ten targets, which is $N = 31$. Using data c_i from Table 3, the probability Eq. (2) for the attack of Washington, D.C. in 2004 shown in Table 4 becomes

$$\begin{aligned} h_1(c;\lambda) = {} & e^{\lambda(1/(1+29.3))*36} \\ & / \\ & (e^{(\lambda(1/(1+47))*413)} + e^{(\lambda(1/(1+34.1))*115)} + e^{(\lambda(1/(1+26.5))*57)} \\ & + e^{(\lambda(1/(1+29.3))*36)} + e^{(\lambda(1/(1+40.4))*34)} + e^{(\lambda(1/(1+23.1))*21)} \\ & + e^{(\lambda(1/(1+19.1))*18)} + e^{(\lambda(1/(1+20))*11)} + e^{(\lambda(1/(1+15.1))*7.3)} \\ & + e^{(\lambda(1/(1+16.5))*6.7)}) \end{aligned}$$

Similarly, by fitting the data, we can get the attack probabilities for all the 31 attacks. By multiplying the probabilities according to Eq. (14), and then logging the likelihood function according to Eq. (15), Fig. 2 shows the maximum value of $l(\lambda)$ is achieved when $\hat{\lambda} = 0.0317$, and we use this values as the baseline for λ.

5. Numerical illustration

In this section, we illustrate Propositions 1 and 3 numerically. Section 5.1 shows the sensitivity analysis on the rationality parameter λ, and Section 5.2 shows the sensitivity analysis on the budget C.

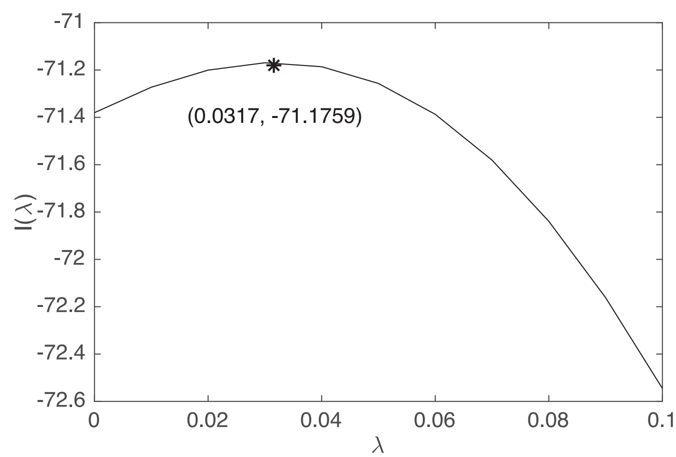

Fig. 2. Optimal value of the attacker's rationality parameter λ.

5.1. Impact of the attacker's rationality parameter λ

We first explore how the attacker's rationality parameter λ affects the optimal allocations. Here, we use the total budget as $271 million, which is the sum of the budget for ten urban areas in 2004 as shown in Table 3. Figs. 3(a, b) illustrate the percentage defensive resource allocation of the ten urban areas when considering pure quantal response and pure game theoretic response, respectively. We observe that the optimal allocation to the area with the relatively high valuation (e.g., New York City, Chicago) will increase when the attacker becomes more rational, while the allocation to the area with the relatively low valuation (e.g., Seattle-Bellevue, Newark) will decrease. The area with higher valuation would receive more investment. As predicted by Proposition 1, the resource allocation in Figs. 3(a) converges to the ones in Fig. 3(b) as λ approach to infinity.

Fig. 4 illustrates the changes of the expected loss with the growth of

Table 4
Terrorism attack frequencies for the top ten urban areas in FY2004-2014.

Urban Area	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
New York City	0	#2	0	#7	#10	0	#13	0	#17,18,19,20	#22	#28,29,30
Chicago	0	0	0	0	0	0	0	0	0	0	0
San Francisco	0	0	0	0	0	0	0	0	0	0	0
Washington, D.C.	#1	0	0	0	0	#12	0	#15,16	#21	#23,24	0
Los Angeles	0	#3	#4,5	#8,9	#11	0	#14	0	0	#25	0
Philadelphia	0	0	0	0	0	0	0	0	0	0	0
Boston	0	0	0	0	0	0	0	0	0	#26	0
Houston	0	0	0	0	0	0	0	0	0	0	0
Newark	0	0	0	0	0	0	0	0	0	0	0
Seattle-Bellevue	0	0	#6	0	0	0	0	0	0	#27	#31
Total	1	2	3	3	2	1	2	2	5	6	4

Source: [49].

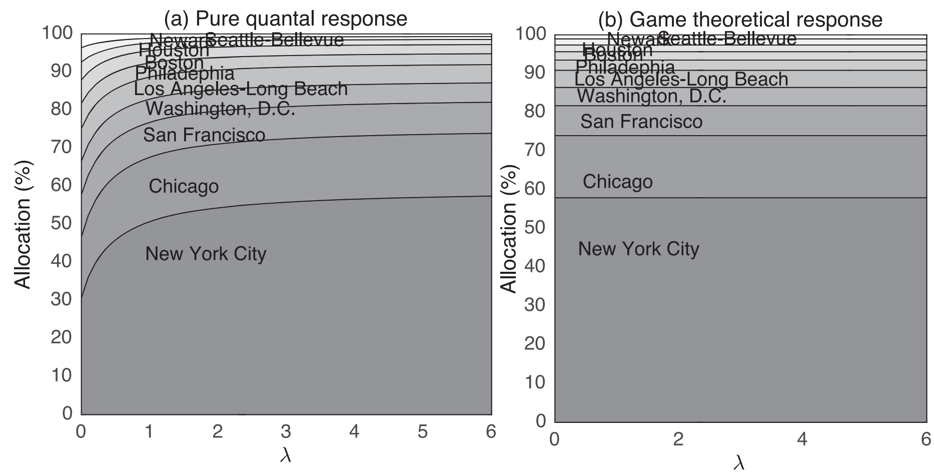

Fig. 3. Optimal budget allocation for both (a) quantal response and (b) game theoretic models as a function of the rationality parameter λ.

λ. When λ increases, the loss for pure quantal response grows with diminishing growth rate. From both Figs. 3(b) and 4, we observe that for the pure game theoretic response model, the rationality parameter λ has no impact on resource allocation and therefore, the expected damage. For both the game theoretic and the quantal response models, we observe that the area with higher valuation gets more resources.

Fig. 4 presents all the four models described in Section 2. Similar to the resource allocation, with the increase of rationality parameter λ, the expected loss from the pure quantal response model approaches to the expected loss from the pure game theoretic model as Proposition 1 shows. This also conforms that when λ increases to infinity, the attacker is perfectly rational. The quantal response false belief model always leads to the highest expected loss, while such loss decreases and approaches to the pure game theoretic response model when λ increases. Therefore, considering quantal response to allocate budget is not always the best strategy for defenders. When the attacker uses game theoretic response, the defender would suffer more losses. The game theoretic false belief and pure game theoretic models lead to similar expected losses. However, game theoretic belief is not always the optimal choice when the attacker uses quantal response. When the attacker is irrational, defenders will suffer less loss by using quantal response strategy.

5.2. Impacts of defense budget C

Next, we explore how the total budget C affects the optimal allocations. As shown in Fig. 4(b), no matter in which model, the expected loss decreases with the increase of total budget, as predicted by Proposition 3. This is because when the total budget increases, each area would have higher allocation and be protected better. For each area, the attack success probability will reduce; thus the total expected loss will decrease. However, no matter how much the budget increases, the rank of the four scenarios in expected loss stays the same.

Table 5 shows the comparison of the game theoretic model results with Proposition 2. When $C = 1000$, $c_i > 0$, all the targets have the same expected loss $p_i v_i$, which equals to L_{GG}. The attacker attacks all the targets randomly; when $C = 10$, only the first three targets receive some allocations. The first two targets have the same expected loss, which is greater than those of other targets. The attacker would randomly attack Target 1 or Target 2.

5.3. Comparison of the allocations with real allocation

We illustrate the models with the data from year 2004 and solve the optimal allocations c^* and c^{**} from quantal response and game theoretic response, respectively. The optimal allocations together with the real allocation are shown in Fig. 5. We see that when compared with the allocation from game theoretic model, the quantal response model gives a closer allocation to that of the real allocation.

6. Conclusion and future research directions

The full rationality assumption about the adversaries adopted by most of the works in defender-attacker games is limited and makes the models sometimes inefficient for application in real world. In this paper, we develop a new defender-attacker model using quantal

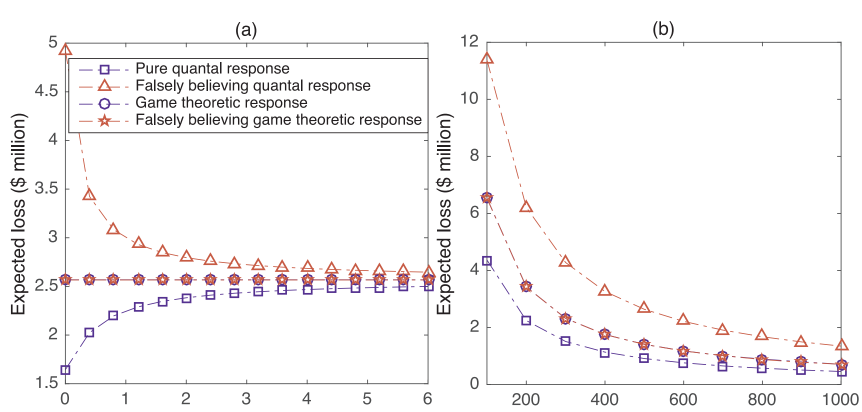

Fig. 4. Expected loss as a function of the attacker's rationality parameter λ, budget C in quantal response model, game theoretic model and other two scenarios described above.

Table 5
Compare game theoretic model results with Proposition 2.

	$C = 1000$, $L_{GG} = 0.71$		$C = 10$, $L_{GG} = 45.00$	
Targets	$p_i v_i$	c_i	$p_i v_i$	c_i
New York City	0.71	579.15	45.00	8.18
Chicago	0.71	160.54	45.00	1.56
San Francisco	0.71	79.06	45.00	0.27
Washington, D.C.	0.71	49.57	36.00	0.00
Los Angeles	0.71	46.76	34.00	0.00
Philadelphia	0.71	28.50	21.00	0.00
Boston	0.71	24.29	18.00	0.00
Houston	0.71	14.45	11.00	0.00
Newark	0.71	9.25	7.30	0.00
Seattle-Bellevue	0.71	8.41	6.70	0.00

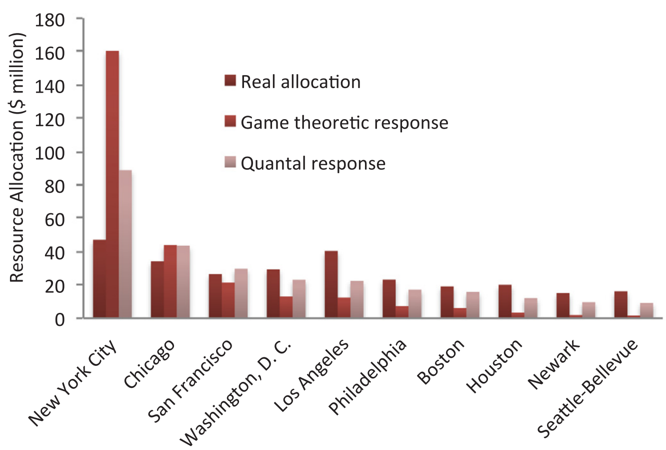

Fig. 5. Defense resource allocation from the game theoretic model and quantal response model, and and compared with the real allocation.

response function by modeling the attacker's rationality with a rationality parameter. When the rationality parameter increases to infinity, the results from quantal response model converges to the game theoretic model. We compare the quantal response model with the game theoretic model. We also consider four possible scenarios regarding the attacker's choices (game theoretic and quantal response) and the defender's beliefs, which are pure quantal response, pure game theoretic response, falsely believing game theoretic response, and falsely believing quantal response. We estimate the attacker's rationality parameter using maximum likelihood technique by adopting real data from multiple data sources. From both the analytical results and the numerical illustration, we find that the expected losses for these four scenarios are generally different. When both the defender and the attacker consider quantal response, the defender suffers the least. However, the defender will have the largest expected loss when the attacker uses game theoretic response and the defender considers the quantal response. In addition, when comparing with the game theoretic response, we find that the quantal response gives a closer allocation to the real allocation.

There exist some future research directions. Besides the logit response function we use in Section 2.2.1, other forms of response function that reflect bounded rationality could be developed and adopted to model the players behavior more realistically. Beyond quantal response, evolutionary game theory could also be used to describe individuals bounded rationality. It would be interesting to study the application of evolutionary game theory in the homeland security domain to address individuals irrationality in making decisions. In addition, this paper only consider the UASI allocation, but other funding sources could also be considered, such as the State Homeland Security Program, the Transportation Security Administration Program, and the Buffer Zone Protection Program.

CRediT authorship contribution statement

Jing Zhang: Formal analysis, Writing - review & editing, Visualization. **Yan Wang:** Writing - original draft. **Jun Zhuang:** Conceptualization, Methodology, Supervision.

Declaration of Competing Interest

The authors declare that they do not have any financial or non-financial conflict of interests.

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Appendix A

Proof for Proposition 1

First, from expressions (1) and (2), it can be shown that the function $L(\mathbf{c})$ is continuous. Moreover, the variables $\mathbf{c}$ are bounded between 0 and C. The set for such variables c_i is compact [50]. According to Extreme Value Theorem, a continuous function reaches a maximum and a minimum on a compact set [37]. Therefore, the function $L(\mathbf{c}, \mathbf{h})$ has an optimal value.

Second, the formula (2) can be transformed as:

$$\begin{aligned} h_j(\mathbf{c}) &= \frac{1}{\frac{\exp(\lambda p_1(c_1)v_1)}{\exp(\lambda p_j(c_j)v_j)} + \ldots + \frac{\exp(\lambda p_{j-1}(c_{j-1})v_{j-1})}{\exp(\lambda p_j(c_j)v_j)} + 1 + \frac{\exp(\lambda p_{j+1}(c_{j+1})v_{j+1})}{\exp(\lambda p_j(c_j)v_j)} + \frac{\exp(\lambda p_n(c_n)v_n)}{\exp(\lambda p_j(c_j)v_j)}} \\ &= \frac{1}{\sum_{i=1}^{n, i\neq j} \exp(\lambda(p_i(c_i)v_i - p_j(c_j)v_j)) + 1} \end{aligned}$$

1. Let j be the index, such that $p_j v_j = \max p_i v_i$. When λ is positive infinite and $p_i v_i < p_j v_j$, then $\exp(\lambda(p_i(c_i)v_i - p_j(c_j)v_j)) = 0$. Donating $|J|$ as the total number of such index j, then $h_j(c) = \frac{1}{|J|}$, $(|J| = 1, 2, \ldots, n)$. As mentioned in Equation (6), the best response for attacking is as follow:

$$\hat{h}_i(\mathbf{c}) = \begin{cases} \frac{1}{|J|}, & \text{if } i \in J \\ 0, & \text{if } i \notin J \end{cases}$$

where $J = \{j: p_j(c_j)v_j = \max p_i(c_i)v_i\}$ refers to the set of targets with the highest expected damage.

2. Let j be the index, such that $p_j v_j < \max p_i v_i$. Assuming there is another index k so that $p_k v_k$ is the maximum, when λ approaches to infinity, $\exp(\lambda(p_k(c_k)v_k - p_j(c_j)v_j))$ is also infinity. Then, we have $h_j(\mathbf{c}) = 0$.

Therefore, when λ is positive infinite, quantal response model becomes identical equivalent to the game theoretic response model.

Proof for Proposition 2

There are three parts to be proved in this proposition when considering game theoretic model to allocate budget. a) The optimal allocation satisfies $p_1v_1 = p_2v_2 = ...=p_nv_n$. b) The optimal allocation is $c_i = \frac{v_i(na+C)}{\sum_{i=1}^n v_i} - a$. c) No matter what response the attacker uses, the total expected loss will always equal to $p_i(c_i)v_i$.

When using game theoretic model, the expected loss formula can be simplified as $\min_c \max_h p_i v_i$, where $p_i = \frac{a}{a+c_i}$. The three parts can be proved respectively below. a) There exists an area k where $p_k v_k < \max p_i v_i$, k's allocation c_k can decrease to increase $p_k v_k$. When the total budget to allocate is specified and certain, the other areas, except k, will have more budget for allocation, and their pv will decrease. The maximum value of pv will also reduce. Therefore, the equilibrium is when the value of pv for all areas are equal, which will also result in the minimum total expected loss. b) When $p_1v_1 = p_2v_2 = ...=p_nv_n = \frac{1}{m}$, we have $\frac{av_i}{a+c_i} = \frac{1}{m}$ and $mav_i = a + c_i$. Then, adding n areas together, we have $ma\sum_{i=1}^n v_i = na + \sum_{i=1}^n c_i$, where $\sum_{i=1}^n c_i = C$. We get $\frac{1}{m} = \frac{a\sum_{i=1}^n v_i}{na+C} = \frac{av_i}{a+c_i}$; Therefore, we have $c_i = \frac{v_i(na+C)}{\sum_{i=1}^n v_i} - a$. c) In both pure game theoretic response and game theoretic false belief, they satisfy $p_1v_1 = p_2v_2 = ...=p_nv_n$, $h_i = \frac{1}{n}$. In both scenarios, the values of total budget C and areas' valuations are the same. Thus, we let $p_iv_i = pv$. In pure game theoretic response, total expected loss is equal to $\sum_{i=1}^n \frac{p_iv_i}{n} = pv$.

In game theoretic false belief, the total expected loss: $L = \sum_{i=1}^n pv\frac{\exp(\lambda p_i(c_i)v_i)}{\sum_{j=1}^n \exp(\lambda p_i(c_i)v_i)} = pv$. In both scenarios, the total expected loss are the same, pv.

Proof for Proposition 3

To consider the change of total expected loss with the increase total budget C, we firstly look at each area's expected loss. According to the Equations (1), (2) and (6), for the target i, its expected loss is $p_i(c_i)v_ih_i$, where $p_i(c_i) = \frac{a}{a+c_i}$ and h_i can be a constant or satisfy $h_i = \frac{\exp(\lambda p_i(c_i)v_i)}{\sum_{j=1}^n \exp(\lambda p_i(c_i)v_i)}$. When the total budget increases, c_i will increase, and p_i will decrease.

1) When h_i is a constant, when p_i decreases, the expected loss $p_i(c_i)v_ih_i$ decreases.

2) When $h_i = \frac{\exp(\lambda p_i(c_i)v_i)}{\sum_{j=1}^n \exp(\lambda p_i(c_i)v_i)}$, taking partial derivative, we have

$$\frac{\partial h_j}{\partial c_j} = -\frac{\frac{\lambda a v_j}{(a+c_j)^2}\exp(\frac{\lambda a v_j}{a+c_j})\sum_{i=1}^{n,i\neq j}\exp(\frac{\lambda a v_i}{a+c_i})}{\left(\sum_{i=1}^n \exp(\frac{\lambda a v_i}{a+c_i})\right)^2} < 0$$

.

Thus, when c_j increases, h_j decreases. The decrease for both p_j and h_j will result in the reduction of expected loss for target j. Furthermore, the total expected loss of the models will decrease due to the reduction of the loss in each target.

References

[1] U.S. Government Accountability Office. DHS risk-based grant methodology is reasonable, but current version's measure of vulnerability is limited. 2008. Accessed in July, 2020; URL http://www.gao.gov/new.items/d08852.pdf.
[2] DHS. Dhs budget. 2020. Accessed in July, 2020; URL https://www.dhs.gov/dhs-budget.
[3] Cox LA. Game theory and risk analysis. Risk Anal 2009;29(8):1062–8.
[4] Zhang J., Zhuang J. Validation, verification, and uncertainty quantification for models with intelligent adversaries. 2017. Handbook of Uncertainty Quantification, R. Ghanem, D. Higdon and H. Owhadi (eds.), Springer, pp. 1–19, in press.
[5] Song C, Zhuang J. Two-stage security screening strategies in the face of strategic applicants, congestions and screening errors. Ann Oper Res 2017;258(2):237–62.
[6] Zhang J, Zhuang J, Behlendorf B. Stochastic shortest path network interdiction with a case study of arizona-mexico border. Reliab Eng Syst Saf 2018;179:62–73.
[7] Levitin G, Hausken K. False targets efficiency in defense strategy. Eur J Oper Res 2009;194:155–62.
[8] Golalikhani M, Zhuang J. Modeling arbitrary layers of continuous level defenses in facing with a strategic attacker. Risk Anal 2011;31(4):533–47.
[9] Levitin G, Hausken K. Defence and attack of systems with variable attacker system structure detection probability. J Oper Res Soc 2017;61(1):124–33.
[10] Levitin G, Hausken K, Yuanshun D. Optimal defense with variable number of overarching and individual protections. Reliab Eng Syst Saf 2014;123:81–90.
[11] Hausken K, Zhuang J. Defending against a terrorist who accumulates resources. Mil Oper Res 2011;16(1):21–39.
[12] Hausken K, Zhuang J. The timing and deterrence of terrorist attacks due to exogenous dynamics. J Oper Res Soc 2012;63(6):726–35.
[13] Hausken K, He F. On the effectiveness of security countermeasures for critical infrastructures. Risk Anal 2016;36(4):711–26.
[14] Shan X, Zhuang J. Modeling cumulative defensive resource allocation against a strategic attacker in a multi-period multi-target sequential game. Relia. Engine. Sys. Saf. 2018;179:12–26. Games and Decisions in Reliability and Risk, doi:10.1016/j.ress.2017.03.022.
[15] Zhang J, Zhuang J. Modeling a multi-target attacker-defender game with multiple attack types. Reliab Eng Syst Saf 2019;185:465–75.
[16] Ben Yaghlane A, Azaiez M, M M. System survivability in the context of interdiction networks. Reliab Eng Syst Saf 2019;185:362–71.
[17] Shan XG, Zhuang J. A game-theoretic approach to modeling attacks and defenses of smart grids at three levels. Reliab Eng Syst Saf 2020;195:106683. https://doi.org/10.1016/j.ress.2019.106683.
[18] Peng R, Xiao H, Guo J, Lin C. Defending a parallel system against a strategic attacker with redundancy, protection and disinformation. Reliab Eng Syst Saf 2020;193.
[19] Zhai Q, Peng R, Zhuang J. Defenderattacker games with asymmetric player utilities. Risk Anal 2020;40(2):408–20. https://doi.org/10.1111/risa.13399.
[20] Aziz R., He M., Zhuang J. An attacker-defender resource allocation game with substitution and complementary effects. Risk Analforthcoming.
[21] Xu Z, Zhuang J. A study on a sequential one-defender-n-attacker game. Risk Anal 2019;39(6):1414–32. https://doi.org/10.1111/risa.13257.
[22] Hausken K. Risk, production, and conflict when utilities are as if certain. Int J Decis Sci, Risk Manage 2010;2(3/4):228–51.
[23] Guan P, Zhang J, Payyappalli V, Zhuang J. Modeling and validating public-private partnerships in disaster management-a sequential game considering risk preferences. Decis Anal 2018;15(2):55–71.
[24] Jose VRRJZ. Incorporating risk preferences in stochastic noncooperative games. IISE Transactions 2018;50(1). 1–1
[25] Zhang J, Zhuang J, Jose VRR. The role of risk preferences in a multi-target attacker-defender resource allocation game. Reliab Eng Syst Saf 2018;169:95–104.
[26] Wu D, Yan X, Peng R, S.Wu. Risk-attitude-based defense strategy considering proactive strike, preventive strike and imperfect false targets. Reliab Eng Syst Saf 2020;196.
[27] Hall JR. The elephant in the room is called game theory. Risk Analysis 2009;29(8). 1061–1061
[28] Loomes G, Sugden R. Regret theory: an alternative theory of rational choice under uncertainty. Econ J 1982;92(368):805–24.

[29] Lopes LL. Between hope and fear: the psychology of risk. In: Berkowitz L, editor. Advances in Experimental Social Psychology. 20. San Diego, CA: Academic Press; 1987. p. 255–95.
[30] Lopes LL, Oden GC. The role of aspiration level in risky choice: acomparison of cumulative prospect theory and SP/a theory. J Math Psychol 1999;43(2):286–313.
[31] Kahneman D, Tversky A. Prospect theory: an analysis of decision under risk. Econometrica 1979;47(2):263–91.
[32] Tversky A, Kahneman D. Advances in prospect theory: cumulative representation of uncertainty. J Risk Uncertain 1992;5(4):297–323.
[33] McKelvey DR, Palfrey RT. Quantal response equilibria for normal form games. Games Econ Behav 1995;10(1):6–38.
[34] Jessie DT, Saari DG. From the Luce choice axiom to the quantal response equilibrium. Journal of Mathematical Psychology 2015. Doi:10.1016/j.jmp.2015.10.001
[35] Ashford AR, Smith CS. General models for quantal response to the joint action of a mixture of drugs. Biometrika 1964;51(3–4):413–28.
[36] Goeree KJ, Holt AC, Palfrey RT. Quantal response equilibrium and overbidding in private-value auction. J Econ Theory 2002;104(1):247–72.
[37] Owen JR. A bayesian sequential procedure for quantal response in the context of adaptive mental testing. J Am Stat Assoc 1975;70(350):351–6.
[38] Portier CJ, Bailer AJ. Testing for increased carcinogenicity using survival-adjusted quantal response test. Fundam Appl Toxicol 1989;12(4):731–7.
[39] Pita J, Jain M, Ordonez F, Tambe M, Kraus S. Solving stackelberg games in the real-world: addressing bounded rationality and limited observations in human preference models. Artif Intell 2010;174(15):1142–71.
[40] Yang R, Kiekintveld C, Ordones F, Tambe M, John R. Improving resource allocation strategies against human adversaries in security games: an extended study. Artif Intell 2013;195:440–69.
[41] Bier VM, Oliveros S, Samuelson L. Choosing what to protect: strategic defensive allocation against an unknown attacker. J Public Econ Theory 2007;9(4):563–87.
[42] Jose VRR, Zhuang J. Technology adoption, accumulation, and competition in multi-period attacker-defender games. Military Operations Research 2013;18(2):33–47.
[43] Shan X, Zhuang J. Modeling cumulative defense resource allocation against a strategic attacker in a multi-period multi-target game. Reliab Eng Syst Saf 2018;179:12–26.
[44] Goeree J, Holt C, Palfrey T. Quantal response equilibrium: a Stochastic theory of games. Princeton University Press; 2016.
[45] Shan X, Zhuang J. Hybrid defensive resource allocation in the face of partially strategic attackers in a sequential defender-attacker game. Eur J Oper Res 2013;228(1):262–72.
[46] DHS. 2004 Urban Area Security Initiative (UASI) allocations. 2004. Accessed in July, 2020; URL https://epic.org/privacy/surveillance/spotlight/0505/stgrant04.pdf.
[47] Willis HH, Morral AR, Kelly TK, Medbly JJ. Estimating terrorism risk. RAND Corporation; 2005. Available at http://www.rand.org/content/dam/rand/pubs/monographs/2005/RAND_MG388.pdf Accessed in July, 2020
[48] DHS. FY 2016 homeland security grant program. 2016. Accessed in July, 2020; https://www.fema.gov/fiscal-year-2016-homeland-security-grant-program.
[49] START. Global terrorism database [data file]. 2016. Retrieved from http://www.start.umd.edu/gtd Accessed in July, 2020.
[50] Wachsmuth G.B. Interactive real analysis. 2007. Accessed in July, 2020; http://pirate.shu.edu/~wachsmut/ira/topo/compact.html#cpctbdd.